SAI SANTOSH CHEETHIRALA

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Engineer specializing in quality, reliability, and the harnesses that verify other systems. Found 56 acknowledged writes lost in a shipping database, caught a split-brain consensus bug through seeded fault injection, and gated a 7.2K-LOC service behind 135 automated tests.

EXPERIENCE

PESU Intelligence Labs — Research Intern, RAG & LLM Reliability **Bengaluru | Jun 2026 – Jul 2026**

No Single Culprit — Causal Attribution Benchmark & Evaluation Harness for Multi-Hop RAG **github.com/santoshcheethiralame-dot/LINEUP**

Python, pytest, GitHub Actions (CI/CD), REST API, Streamlit, vLLM, BM25, PyTorch, HuggingFace Transformers, Conformal Prediction

- Built and shipped a 7.2K-LOC evaluation service — public REST API, pooled leaderboard, and Streamlit explorer — with 135 automated tests gating every change on GitHub Actions CI
- Designed a non-circular counterfactual oracle producing leak-free ground-truth labels without ever consulting the method under test, replacing an evaluation loop that had been silently circular
- Ran ~100K teacher-forced inferences across open-weight models (7B-120B) served with vLLM against a live BM25 retriever, sweeping three multi-hop datasets and retrieval depths k=20-50

PROJECTS

highwater — Jepsen-Style Durability Analysis of Couchbase **github.com/santoshcheethiralame-dot/HighWater**

Go, Couchbase DCP, Porcupine linearizability checker, gocbcore patch, fault injection

- Ran a Jepsen-style durability analysis of Couchbase — one of the few major distributed databases with no published Jepsen report — showing 56 acknowledged writes lost (50 regressed, 6 absent) at durability level "majority" when the in-memory majority is killed before the disk flush completes, with the history proven non-linearizable
- Built a white-box harness that taps the DCP stream to observe prepare/commit/abort directly, rather than inferring from client responses, so "never prepared" and "prepared then aborted" are distinguishable; required patching gocbcore to expose the stream
- Confirmed the result under two independent methods that agree on both arms — a direct value comparison and Porcupine, a general linearizability checker with no knowledge of durability levels — and showed the control arm (persistToMajority) loses zero writes and stays linearizable

tessera — Distributed, Sharded, Replicated Key/Value Store (Rust, std-only) **github.com/santoshcheethiralame-dot/TESSERA**

Rust (0 external crates), Raft, LSM-tree, Deterministic Simulation Testing, Linearizability Checking, MVCC

- Engineered deterministic simulation testing: time, network, disk, scheduling, and randomness are injectable and the whole cluster runs single-threaded in a seeded world, so failure interleavings (partitions, message loss/duplication, reordering, crashes) are fuzzed by the thousand and replay exactly from a seed (TigerBeetle/FoundationDB approach)
- Caught a real split-brain bug via seed 99 (a duplicated vote reply let a candidate win a sub-majority, producing two leaders in one term); a Wing-Gong linearizability checker validates client histories across 1,000 randomized failure schedules with zero violations

Gradient — GPU-Native Deep-Learning Framework, In-Browser **santoshcheethiralame-dot.github.io/GRADIENT**

TypeScript, WebGPU, WGSL, React, Web Workers

- Validated every gradient three ways (f64 CPU oracle 51/51, finite-difference, PyTorch parity to ~1e-7) and profiled a tiled matmul to ~3-4x a naive kernel

TECHNICAL SKILLS

Testing & Reliability: Deterministic simulation testing, randomized fault injection, linearizability checking, regression testing, pytest, CI/CD (GitHub Actions), model evaluation

Languages: Python, C, Rust, TypeScript, JavaScript, SQL, WGSL

Systems & Backend: Distributed systems (Raft, LSM storage, MVCC), FastAPI, REST APIs, Docker, Linux, WebGPU/WGSL, vLLM

Tools: Git, GitHub Actions, Linux, Hugging Face Spaces, GPU compute, numerical gradient checking

EDUCATION

PES University — B.Tech, Computer Science Engineering **Bengaluru | Sep 2024 – May 2028 (Expected)**

CGPA 7.95/10 (through Sem 4) — CNR Merit Scholarship (top 25% cohort), Sem 1 — Academic Distinction Award (SGPA 7.75+), Sem 2 & Sem 4

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, NLP, Linear Algebra, Data Science

LEADERSHIP & ACTIVITIES

Sahay AI — Student AI Club, PES University — Multimedia Domain Head **Oct 2025 – Mar 2026**

- Built a content pipeline and posting schedule for a 50+ member AI club, driving 5-6x growth in content views over three months
- National-Level Air Rifle Competitor - 10m Air Rifle, National School Games (a precision event built on sustained focus)